EE-M-002142A(Army-GL)
April 30, 1986
SUPERSEDING
EE-M-002142(Army-GL)
March 15, 1983

INTERIM FEDERAL SPECIFICATION

MACARONI AND CHEESE, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the US Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers macaroni and cheese thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of the Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

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U.S. Department of Agriculture (USDA)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Butter

(Application for copies should be addressed to the Dairy Standardization Section, Diary Division Room 2750-S, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

American Dehydrated Onion and Garlic Association (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, P.O. Box 2109, Novato, CA 94947.)

National Academy of Science

Food Chemicals Codex

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(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Macaroni, elbow. Elbow macaroni shall be made with enriched Durum Semolina Wheat Flour. The outer curvature length shall range between 0.75 and 1.0 inches with a diameter ranging from 0.16 to 0.2 inches. The wall thickness shall range between 0.034 and 0.048 inches. The macaroni shall contain glycerol monostearate. Moisture content shall not exceed 13 percent.

3.2.2 Water. Water used for formulation, blanching, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.3 Cheese, processed, pasteurized, american, white. The pasteurized processed american cheese shall conform to the Food, Drug and Cosmetic Act Standards of Identity for pasteurized processed cheese except the cheese shall have a maximum of 40 percent moisture and a maximum of 2.5 percent salt and not less than 50 percent fat. The cheese shall possess a mild and desirable pasteurized processed cheese flavor with no more than slight emulsifier flavor. But shall be free from bitter, sour, flat, yeasty or moldy flavors. The body shall be smooth, medium firm, and resilient. The cheese shall be white in color, in excellent condition, and free from mold, slime, discoloration and burnt particles.

3.2.4 Butter, lightly salted. The lightly salted butter shall be U.S. Grade A or better of the U.S. Standards for Grades of Butter.

3.2.5 Cheese, dehydrated cheddar. Dehydrated cheese shall be derived from extra sharp, aged cheddar cheese. The dehydrated cheese shall be uniform in composition and appearance, free from lumps that do not fall apart under light finger pressure, and free from burnt specks and foreign material. The cheese shall be free from rancid, stale, oxidized, tallowy, bitter, fishy, soapy, scorched or other objectionable flavors and odors. The flavor shall be that of a typical sharp-cured processed cheddar cheese. The dehydrated cheese shall contain not more than 3.0 percent moisture, 5.0 percent salt and not less than 48.0 percent milkfat.

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3.2.6 Starch, waxy maize, modified. The starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a viscosity (be fully hydrated) at common retort temperatures. It shall resist breakdown at low pH, under sheer stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.7 Salt. Salt shall be non-iodized, white, refined sodium chloride with or without anti-caking agents.

3.2.8 Sugar, white, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet sugar or a combination thereof and shall comply with MIL-STD-900.

3.2.9 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.10 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.11 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting flavor, and a light color. The white pepper shall contain not less than 1.0 mL volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.12 Mustard flour. Mustard flour shall be a bright yellow powder prepared from a blend of powder derived from the endosperm of the seed of Brassica hirta Moench and Brassica juncea. Mustard flour shall contain not less than 0.4 percent volatile oil and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.13 Guar gum. Guar gum shall conform to the Food Chemicals Codex.

3.2.14 Color, annatto, dry. The annatto color shall be a reddish orange water soluble powder of the extractive of Annatto Bixa Orellana. The powder shall contain less than 5 percent moisture. One hundred percent shall pass through a U.S No. 10 Standard sieve and shall be free from hard lumps. The total color (as percent norbixin) shall be 15 percent.

3.2.15 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

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3.3 Preparation and further processing.

3.3.1 Macaroni preparation. The macaroni shall be blanched in boiling water until the weight increases by 2.5 times. The macaroni shall be immediately cooled to less than 90°F and rinsed to remove excess starch. If the macaroni is not immediately filled into tray pack cans, it shall be cooled to 40°F. The macaroni shall be placed in trays within 2 hours after blanching has been accomplished.

3.3.2 Sauce preparation. The sauce shall be formulated as follows:

Ingredient	Percent
Water	69.290
Cheese, processed, pasteurized, american, white	18.400
Butter, Lightly salted	4.600
Cheese, dehydrated, cheddar	4.000
Starch, waxy maize modified	1.500
Salt	1.000
Sugar, white	.550
Lecithin	.253
Onion powder	.250
Pepper, white	.080
Mustard, flour	.050
Guar gum	.020
Annatto, dry	.007
	100.00

a. Add butter, cheeses, and approximately 2/3 of the water to the processing vessel. Heat with agitation to 180°F.

b. Add salt, sugar, lecithin, onion powder, pepper, and mustard flour to the mixture with constant agitation.

c. Blend guar gum into approximately 1/2 of the reserved water and add this with constant agitation.

d. Prepare a thin starch slurry using the remainder of the water and add slowly with constant agitation.

e. Heat entire contents to 185°F and hold for 5 minutes.

f. Add color and blend until color of sauce is uniform.

g. The volume of the final mixture shall be adjusted to compensate for evaporation loss during heating.

h. The sauce shall be filled within 4 hours after cooking and hold at a temperature of 150° to 180°F during this interval.

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3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with blanched macaroni and sauce such as to conform to the finished product requirements and to the following requirements:

a. The temperature of the sauce at time of filling shall be 150°F minimum.

b. The temperature of the macaroni at time of filling shall be not greater than 40°F unless the filling occurs immediately after blanching.

c. The cans shall be hermetically sealed with a minimum of 10 inches (HG) vacuum immediately after filling.

d. The filled and sealed cans shall be in the retort process within 1 hour after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 240°F until a sterilization value (F_o) of not less than 6.0 has been achieved. The filled and sealed thermoprocessed tray pack cans shall show no evidence of can swelling when tested for sterility as specified in 4.6.4.

3.6 Finished product. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. The sauce shall not be thin, thick, lumpy, or oily.

e. The sauce shall have a typical processed cheese flavor.

f. The macaroni shall not be dry, rubbery, mushy, or crumbly.

g. The salt content shall be not less than 0.5 percent nor greater than 1.5 percent.

h. No individual can shall weigh less than 104 ounces.

i. The average net weight shall be not less than 106 ounces.

j. No individual can shall contain less than 60 ounces nor more than 75 ounces of washed drained weight of macaroni.

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3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Production acceptability shall be determined by USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this specification.

4.3 First article inspection. When a first article is required (see 6.1) it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, US Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight or any individual can is less than 104 ounces, the lot shall be rejected.

4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated at least 30 minutes in 160° to 180°F water, opened, and inspected for the defects listed in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8	-	0	2
	8	16	1	2
3201-35000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

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TABLE II. Product inspection 1/ 2/

Category	Defect
Major	
101	Washed and drained weight of macaroni in a can is less than 60 ounces nor more than 75 ounces. 3/
102	Sauce thin, thick, lumpy, or oily.
103	Sauce does not have a typical processed cheese flavor.
104	Macaroni texture dry, rubbery, mushy, or crumbly.

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine washed and drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquify the sauce without breaking the macaroni pieces. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve without breaking the macaroni. Sieve Area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted slightly and allowed to drain at least 2 minutes before determining the washed and drained weight by subtracting the sieve tare from the gross weight. The washed and drained weight shall be reported to the nearest 0.1 ounce.

4.6.3 Salt content testing. Nine tray packs shall be selected at random from the lot and distributed as follows:

-Three for laboratory analysis.
-Three for submission to the contractor.
-Three for retention by an AMS agent as a reserve sample.

The three tray packs for laboratory analysis shall be individually tested for salt content in accordance with the Official Methods of Analysis of the

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Association of Official Analytical Chemists, Chapter: Meat and Meat Products, Section: Meat, except that preparation of the samples shall be as follows. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if the salt content of any individual tray pack is less than 0.5 percent or more than 1.5 percent. Results shall be reported to the nearest 0.1 percent. Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt content requirement. When the reserve samples are analyzed, the analyses for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.6.4 Sterility testing. The sample size shall be one filled, sealed, and thermoprocessed tray pack can selected from each retort batch in the lot and shall be tested for sterility by incubating the cans at 95°F ± 2°F for 10 days. Any evidence of can swelling shall be considered a critical defect and cause rejection of the lot.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: (1) National stock number, item description, contract number, or date of pack markings missing, incorrect or illegible; (2) Not reinforced with nonmetallic strapping or tape; Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (4) Length or width of pads less than box dimensions by more than 1/8 inch; (5) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of a sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can

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shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither effect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3 shall be arranged in unit loads in accordance

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with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray can shall be labeled in accordance with MIL-L-1497. In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge sealed cans in boiling water. Bring to a boil again and continue boiling for 40-45 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. Caution: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of tray can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING

Do not place closed can in oven, as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 18 servings of 2/3 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

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6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8940-A503

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Natick, MA 01760-5014

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
EE-M-002142A(Army-GL)	Macaroni And Cheese, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP-1, Feb 86

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